HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and modified **Sattline control system software** for tank farm operations, implementing process sequences and interlocks based on functional design specifications.
- Participated in **software testing and validation activities** for control system deployments, ensuring adherence to process requirements and quality standards.
- Performed **I/O checks and loop testing** during commissioning phases, verifying signal integrity and system responsiveness in a manufacturing environment.
- Assisted with **Clean-In-Place (CIP) commissioning** and validation testing, ensuring system readiness for production runs in a multinational manufacturing setting.
- Collaborated with process engineers and operations personnel to troubleshoot **automation and control system issues** during commissioning and initial production runs.
- Supported **documentation activities**, including the creation of test records, punch lists, and commissioning reports, contributing to project closure.
- Containerized Java Spring Boot microservices using **Docker** and deployed them to **AWS EC2 instances**, managing configurations via Jenkins pipelines for automated deployments.

SKILLS

Control Systems – Sattline, PLC, DCS, SCADA, Industrial Automation

Programming Languages – Java, Python, SQL, Shell Scripting

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins

Databases – PostgreSQL, MySQL, Oracle

Tools & Methodologies – Jira, Confluence, Git, Agile, Scrum, CI/CD

PROJECTS

Industrial Control System Automation for Manufacturing

- Developed and tested **PLC control logic** for a simulated batch processing unit, focusing on sequence execution and alarm management.
- Integrated **Python scripts** for automated data logging and performance reporting from the control system, reducing manual effort by approximately **30%**.
- Wrote comprehensive unit tests using **JUnit** for Java-based components that interfaced with the control system's data acquisition layer.

Automated Production Monitoring Dashboard

- Engineered a **Python-based ETL pipeline** to ingest and transform production data from multiple sources into a central data store.
- Built interactive **Power BI dashboards** to visualize key operational metrics, enabling stakeholders to monitor real-time production status and identify bottlenecks.
- Implemented **SQL queries** and data models within Power BI to support complex analytical reporting for process optimization initiatives.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)